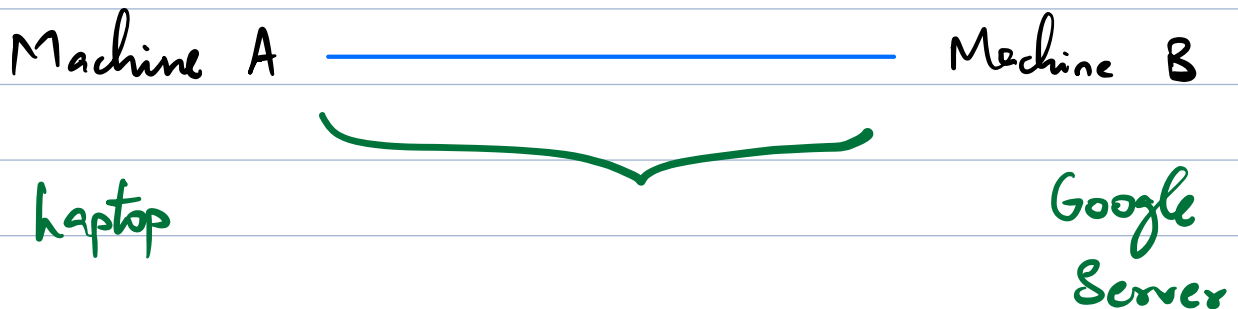
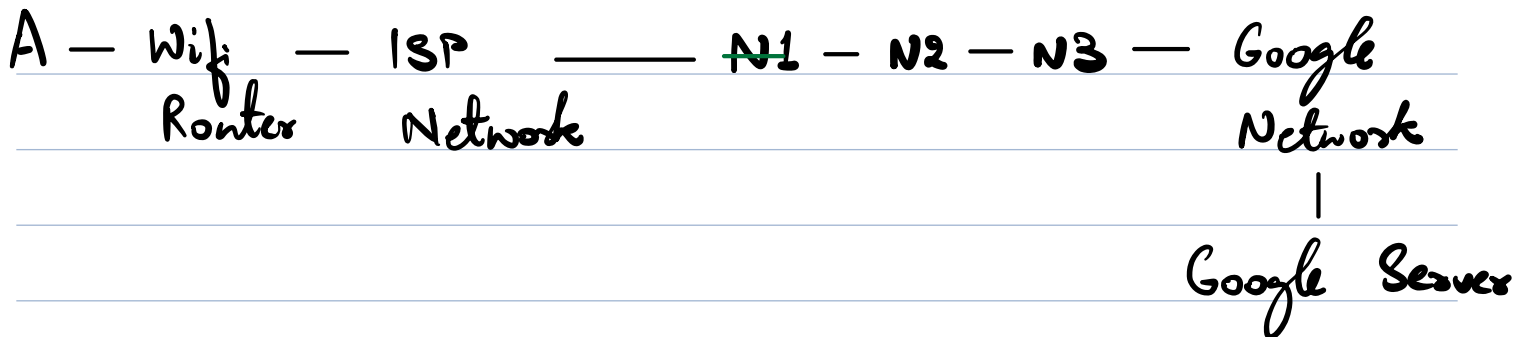


www.google.co.in

- Send a request to Google Server
- DNS
- IP Address
- find google server & send request
- Server returns HTML.



Router



Internet is made from many independently operated networks connected together.



LAN

WIFI

Fiber

Cellular Network

Multiple Intermediate Networks.

Q Why do we need networks ??

Components of A Network

1) End Devices / Hosts

2) Links

3) Intermediate Devices

- Switch

- Router

- Access Points

4) Protocols

Defines rules governing communication b/w entities

HTTP

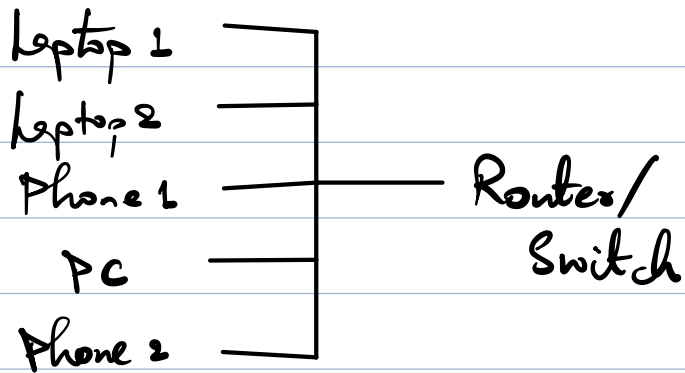
DNS

TCP

UDP
IP

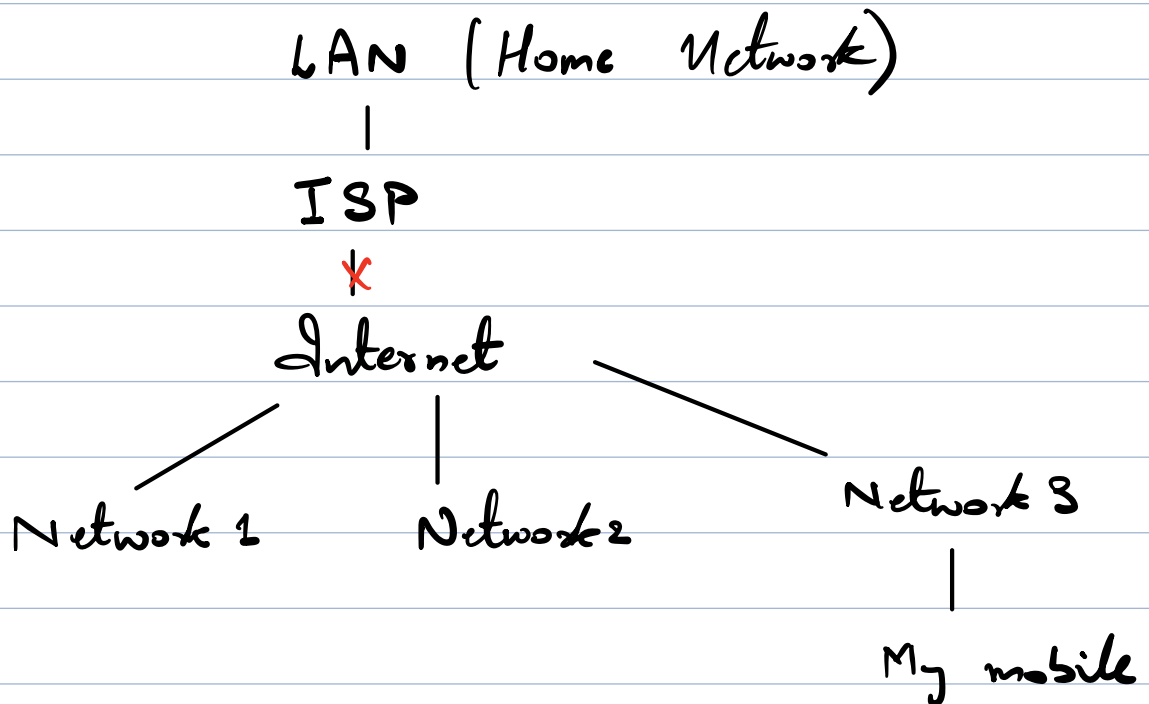
3) Types of Network

LAN - Local Area Network



WAN - Wide Area Network

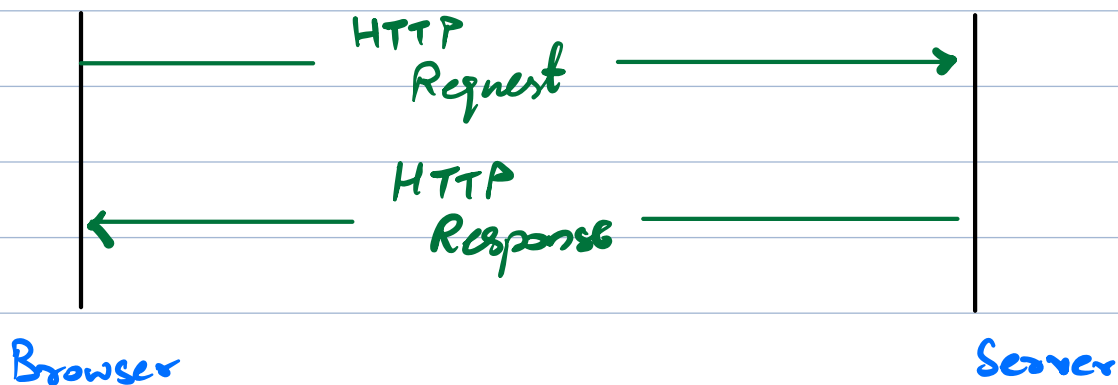
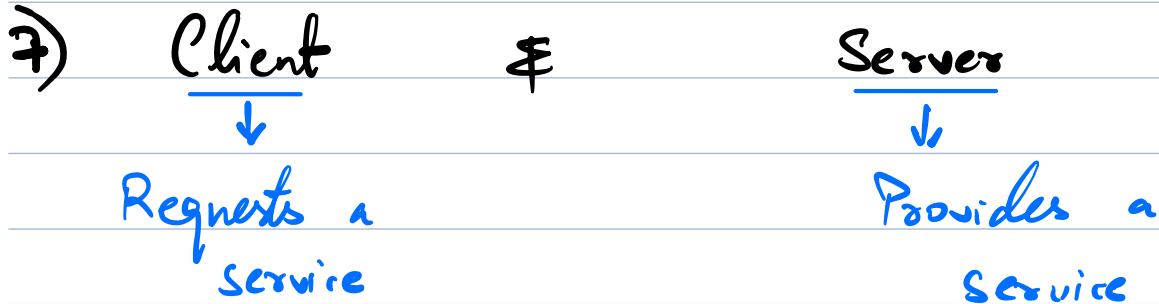
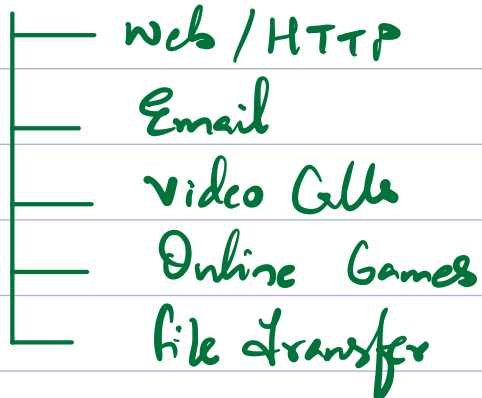
Internet



6) Internet v/s Web (www)

WWW → Application ecosystem running over the internet.

Internet



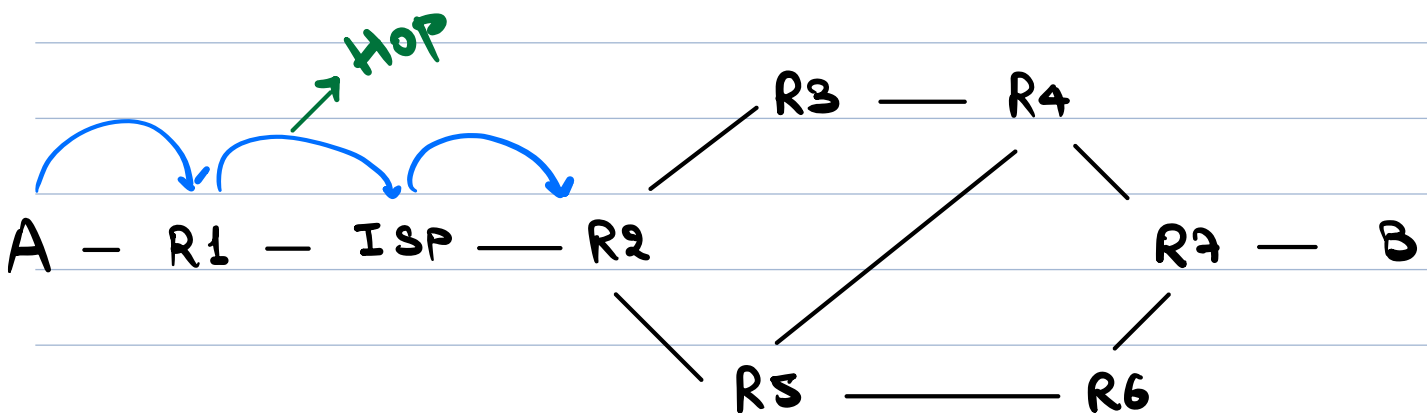
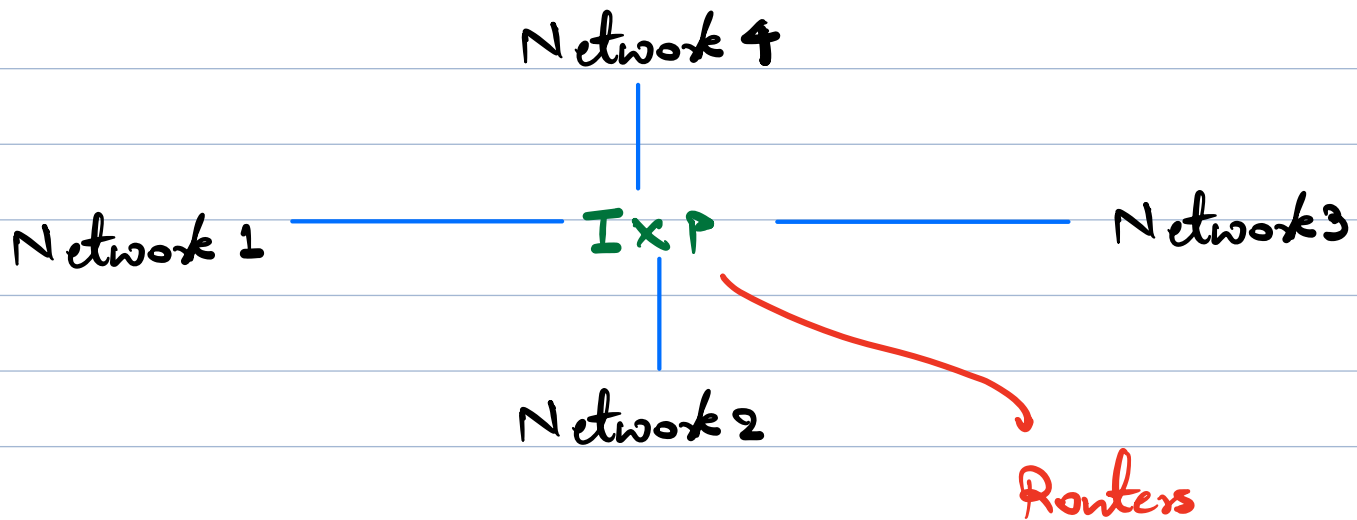
How Networks Connect

ISP

Backbone Network

Internet Exchange Point (IXP)

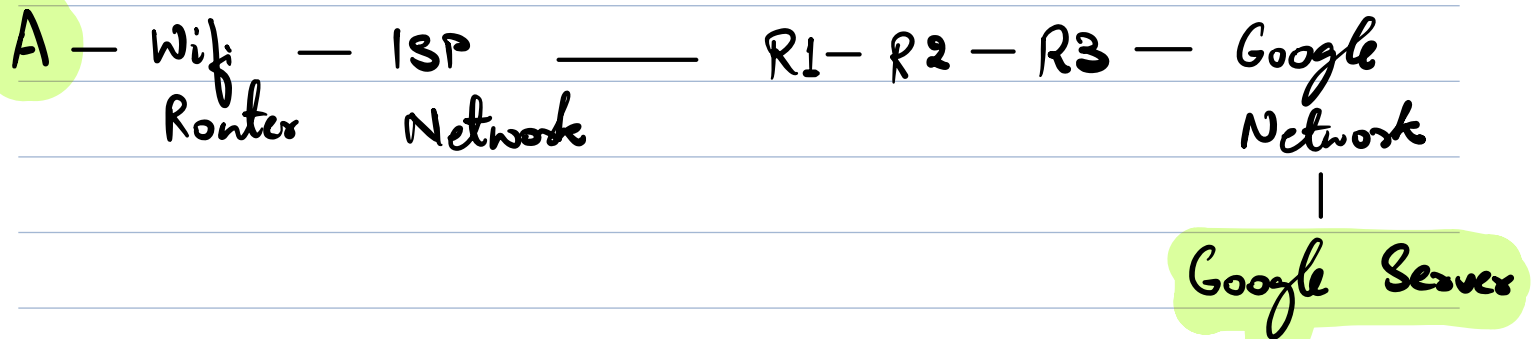
→ Physical Infra/Location where different networks can exchange traffic



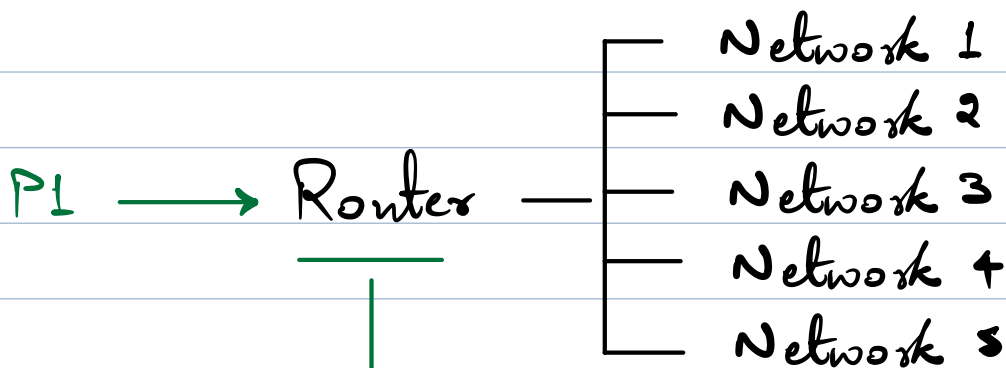
* Routing

* Physical Distance Matters.

Network Edge vs Network Core



Router

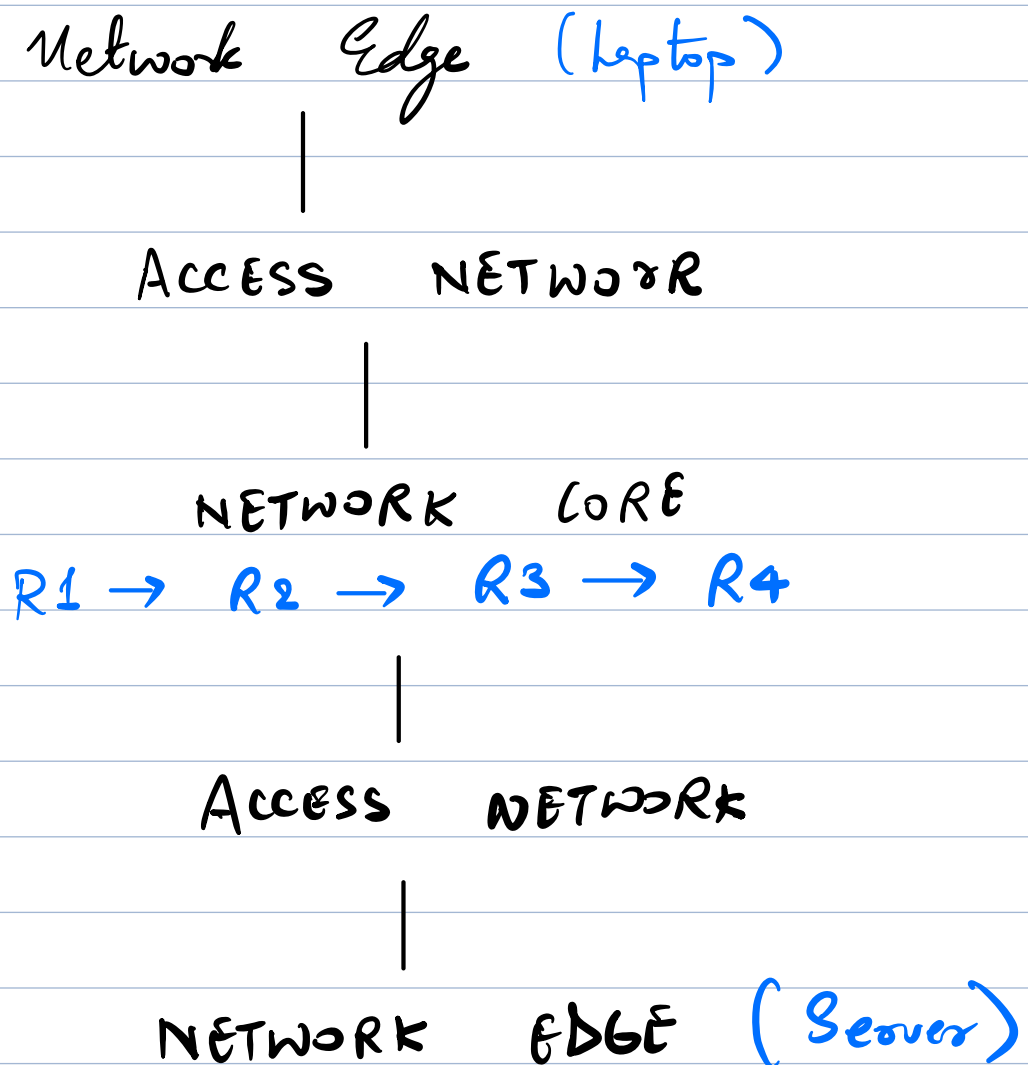


1) forwarding

2) Routing

Router v/s Switch

Access Networks.



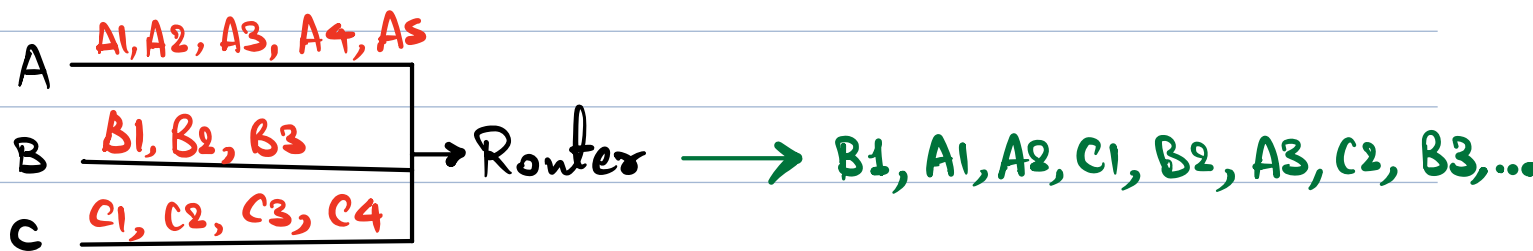
Packet

1) Sharing the network

2) Errors & retransmission.

Header	Payload / Data
--------	----------------

Packet Switching



Statistical Multiplexing.

Store & forward.

Circuit Switching

Resources are reserved for communication session

Network Performance.

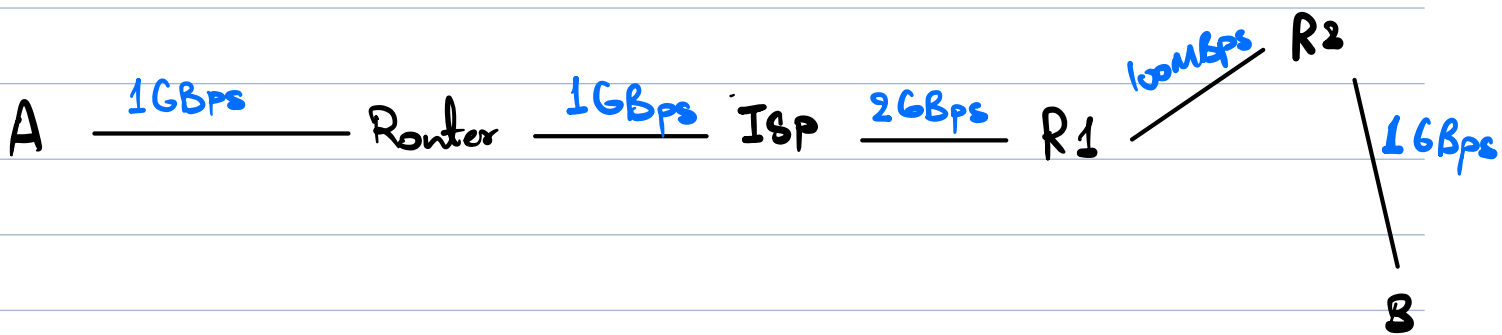
100 MBbs $\longrightarrow$ 1 GBps

1) Bandwidth

Max rate / capacity at which a link can transmit data.

2) Throughput

Actual rate at which data is successfully transferred.



3) Latency

Time taken for data to travel through the network.

Propagation

Transmission

